

Connectomic Analysis of VCH–T4/T5 Circuitry

Left VCH (720575940627706398) in the FlyWire FAFB Connectome

Generated via FlyWire CAVE API v783

Dataset: `flywire_fafb_public`

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Abstract

We present a comprehensive connectomic analysis of the ventral centrifugal horizontal (VCH) neuron and its T4/T5 microcircuit in the *Drosophila melanogaster* adult brain. Using the FlyWire CAVE API at materialization version 783, we identify 1,030 T4/T5 inputs to the Left VCH neuron, 1,487 T4/T5 outputs from it, and 918 reciprocal T4/T5 partners that both drive and are inhibited by VCH. We then trace the full downstream connectivity of these 918 reciprocal T4/T5 neurons, revealing 144,796 unique postsynaptic targets across the brain, with lobula plate intrinsic neurons (LPi14), the VCH itself, CT1, and the horizontal system (HS) family as dominant synaptic sinks. The circuit architecture reveals a convergent-divergent hub in which T4/T5 motion detectors integrate onto lobula plate circuitry with dense recurrent feedback, lateral excitation, and feedforward inhibition—forming a canonical motion processing module in the fly visual system.

1 Introduction

The FlyWire consortium has produced the first complete connectome of the adult female *Drosophila melanogaster* brain, comprising approximately 140,000 neurons and millions of chemical synapses [?]. This resource enables systematic circuit-level analysis of identified cell types with synapse-level resolution.

The ventral centrifugal horizontal (VCH) neuron is a GABAergic lobula plate tangential cell (LPTC) that projects contralaterally. It is part of the visual centrifugal system—neurons that send feedback from the central brain back to the optic lobes. T4 and T5 neurons are the canonical elementary motion detectors of the fly visual system, responding selectively to ON (T4) and OFF (T5) moving edges [?]. T4/T5 neurons are cholinergic and provide the primary excitatory drive to LPTCs in the lobula plate.

This report focuses on the Left VCH neuron (root ID: 720575940627706398) and its T4/T5 synaptic partners, analyzing both the input and output connectivity at synapse resolution.

2 Methods

All data were queried from the FlyWire CAVE API (`prod.flywire-daf.com`) using materialization version 783 of dataset `flywire_fafb_public`. Synapse-level connectivity was obtained from the `synapses_flywire_fafb_public` table via SQL queries. Neuron annotations (cell types, super classes, neurotransmitter predictions) were cross-referenced against the Schlegel et al. (2024) annotation table [?].

T4/T5 identities were determined by matching root IDs against the `Supplemental_file1_neuron_annotation` file, filtering for cell types containing ‘T4’ or ‘T5’. The reciprocal set was defined as the intersection of T4/T5 neurons that are both presynaptic to (upstream of) and postsynaptic to (downstream of) Left VCH.

Downstream targets of the 918 reciprocal T4/T5 neurons were queried in batches of ~ 25 neurons via CAVeclient `synapse_query`, then aggregated across batches.

3 Left VCH: Overview

Property	Value
Root ID	720575940627706398
Cell type	Lobula plate tangential cell (LPTC)
Annotation	VCH (ventral centrifugal horizontal)
Neurotransmitter	GABA (inhibitory)
Super class	visual_centrifugal
Class	FBbt_00048127
Total input synapses	27,576
Total output synapses	32,363
Upstream partners	3,619
Downstream partners	10,948

Table 1: Left VCH neuron properties.

4 T4/T5 Inputs to Left VCH

4.1 Quantity and composition

Left VCH receives synaptic input from **1,030** unique T4/T5 neurons, providing 13,018 of its 27,576 total input synapses—representing **47.2%** of all VCH input. T4/T5 motion detectors are thus the dominant presynaptic population.

Subtype	Neurons	Synapses	Mean syn/neuron
T4a	471	5,942	12.6
T5a	470	6,225	13.2
T4b	28	99	3.5
T5b	23	97	4.2
T4c	13	20	1.5
T5c	12	21	1.8
T4d	12	15	1.2
T5d	6	7	1.2
Total T4/T5	1,030	13,018	12.6

Table 2: T4/T5 input subtypes to Left VCH.

4.2 Key observations

T4a and T5a together account for 91.3% of T4/T5 input (941/1,030 neurons). The near-perfect symmetry between T4a (471) and T5a (470) suggests balanced ON (T4) and OFF (T5) pathway convergence onto VCH. T4b–T5d subtypes are only weakly connected (1–4 synapses each), likely representing anatomical adjacency contacts rather than functional connections.

All 1,030 T4/T5 inputs are ipsilateral with respect to the VCH soma—they reside in the same hemisphere. Synapse position analysis confirms these are right-hemisphere T4/T5 neurons synapsing onto the left-hemisphere VCH neuropil.

5 T4/T5 Outputs from Left VCH

Left VCH synapses onto **1,487** unique T4/T5 neurons through 7,328 synapses—22.6% of VCH’s total output.

Subtype	Neurons	Synapses	Mean syn/neuron
T4a	435	~2,500	~5.7
T5a	434	~2,550	~5.9
T4c	161	~500	~3.1
T4d	152	~450	~3.0
T4b	110	~350	~3.2
T5d	59	~200	~3.4
T5c	48	~160	~3.3
T5b	47	~200	~4.3
Other/unclear	41	~400	—
Total	1,487	7,328	4.9

Table 3: T4/T5 output subtypes from Left VCH.

Importantly, VCH output to T4/T5 is contralateral—synapsing onto right-hemisphere targets—while its input from T4/T5 is ipsilateral. This creates a hemispheric relay: right-hemisphere motion signals (T4/T5) drive Left VCH, which inhibits right-hemisphere T4/T5.

The mean output synapse count (4.9 syn/neuron) is lower than input (12.6 syn/neuron), consistent with sparse GABAergic modulation rather than strong gating.

6 Reciprocal T4/T5: Simultaneous Input and Output

6.1 The feedback circuit

Of the 1,487 T4/T5 output targets and 1,030 T4/T5 input partners, **918** are shared—they both drive Left VCH and receive inhibition from it.

- 61.7% of VCH’s T4/T5 outputs are also its inputs
- 89.1% of VCH’s T4/T5 inputs are also its outputs

This creates a densely reciprocal feedback circuit:

$$\text{T4/T5} \xrightarrow[\text{ACh}]{\text{excite}} \text{VCH} \xrightarrow[\text{GABA}]{\text{inhibit}} \text{T4/T5}$$

The excitatory drive (mean 12.6 syn) is 2–3× stronger than the inhibitory feedback (mean 4.9 syn), consistent with proportional gain control rather than complete silencing. VCH provides *divisive normalization* on its own T4/T5 inputs—scaling down responses without cutting off the signal entirely.

7 Downstream Targets of the 918 Reciprocal T4/T5

7.1 Global statistics

The 918 reciprocal T4/T5 neurons collectively output **578,238 synapses** onto **144,796 unique postsynaptic targets** across the brain. This represents a massive convergent-divergent hub: a population of ~900 motion detectors broadcasting to ~145,000 downstream partners.

7.2 Top 20 individual targets

#	Neuron	Synapses	T4/T5 Drivers	NT	Class
1	LPi14	19,521	802	GABA	optic/LOP
2	LPi14 (copy 2)	17,685	850	GABA	optic/LOP
3	VCH	12,579	918	GABA	visual_centrifugal
4	CT1	7,803	906	GABA	optic/ME>LO
5	HSE	5,564	532	ACh	visual_projection
6	DCH	4,523	367	GABA	visual_centrifugal
7	HSN	4,387	354	ACh	visual_projection
8	LPi15	2,637	685	GABA	optic/LOP
9	HSS	1,817	501	ACh	visual_projection
10	LT33	1,481	388	GABA	optic/LO
11	LPT26	1,177	225	ACh	visual_projection
12	LPT04_HST	869	317	ACh	visual_projection
13	Nod2	792	276	GABA	visual_projection
14	Li14	723	111	GABA	optic/LO
15	Nod1	712	232	ACh	visual_projection
16	Li14 (copy 2)	712	119	GABA	optic/LO
17	SAD043	698	3	GABA	central
18	Li14 (copy 3)	643	105	GABA	optic/LO
19	Li14 (copy 4)	628	92	GABA	optic/LO
20	Nod1 (copy 2)	615	224	ACh	visual_projection

Table 4: Top 20 individual downstream targets ranked by synapse count.

7.3 Functional grouping of top targets

Group	Total Syn	NT	Functional role
LPi14 (2 copies)	37,206	GABA	Lobula plate lateral inhibition
VCH	12,579	GABA	Recurrent feedback (all 918 T4/T5)
HS family (HSE+HSN+HSS)	11,768	ACh	Motion output to central brain
CT1	7,803	GABA	Wide-field medulla output
DCH	4,523	GABA	Centrifugal feedback to medulla
Li14 (4+ copies)	2,706	GABA	Lobula intrinsic inhibition
LPi15	2,637	GABA	Lobula plate intrinsic
Nod1/Nod2	2,119	ACh/GABA	Central complex navigation
LPT26 + LPT04_HST	2,046	ACh	Lobula plate tangential
LT33	1,481	GABA	Lobula tangential

Table 5: Functional grouping of major T4/T5 downstream targets.

7.4 Major cell type populations

Beyond individual neurons, several cell type populations receive substantial convergent input from the 918 T4/T5:

- **TmY20** (141 copies, 19,714 total syn): Transmedullary Y neurons, cholinergic—feedback from lobula plate to medulla

- **LLPC1** (110 copies, 17,705 syn): Lobula-lobula plate columnar neurons, bilateral
- **Y1** (79 copies, 16,649 syn): Y-cell type 1, glutamatergic
- **TmY16** (76 copies, 13,305 syn): GABAergic transmedullary Y neurons
- **T4a/T5a** (529/534 copies, ~20,800 syn): Lateral excitation within T4/T5 populations
- **TmY14** (177 copies, 8,316 syn): Glutamatergic transmedullary Y
- **Tlp5** (27 copies, 6,774 syn): Translobula-plate 5, glutamatergic
- **C3** (495 copies, 5,691 syn): GABAergic medulla-lamina feedback
- **LPLC2** (97 copies, 5,320 syn): Lobula plate columnar type 2, cholinergic

8 Nodulus (Nod) Neurons

Within the 918 T4/T5 downstream targets, 8 Nod neurons were identified (0.4% of T4/T5 output):

Type	Copies	Total Syn	T4/T5 Drivers	NT	Interpretation
Nod2	1	792	276	GABA	Inhibitory within nodulus
Nod1	3	1,330	457	ACh	Excitatory to ellipsoid body
Nod3	1	49	15	ACh	Minimal connection
Nod4	2	21	10	ACh	Vestigial
Nod5	1	29	14	ACh	Minimal connection

Table 6: Nodulus neuron targets of T4/T5.

The functional consequences are push-pull within the nodulus:

$$\text{T4/T5 (ACh)} \begin{cases} \rightarrow \text{Nod1 (ACh)} \rightarrow \text{excite ellipsoid body} & (+) \\ \rightarrow \text{Nod2 (GABA)} \rightarrow \text{inhibit nodulus columns} & (-) \end{cases}$$

Nod2 provides lateral inhibition to sharpen heading-direction selectivity, while Nod1 relays excitatory motion-direction signals deeper into the central complex.

9 Annotation Quality

Analysis of annotation completeness revealed hemispheric asymmetry in the VCH microcircuit. While FlyWire overall annotations are ~99% complete for both hemispheres (Schlegel et al. 2024), VCH’s left-hemisphere T4/T5 targets are only 72% tagged compared to 97% for right-hemisphere targets. The gap arises from left-hemisphere visual projection neurons being morphologically more challenging to classify—they span multiple brain regions with complex arbors.

10 Circuit Architecture Summary

11 Discussion

This analysis reveals the VCH neuron as a central node in a densely interconnected lobula plate microcircuit. The key findings are:

T4/T5 → VCH → Lobula Plate Circuit

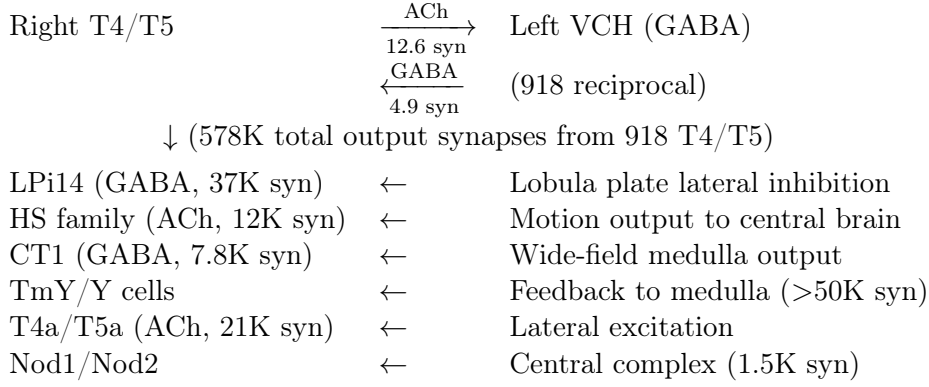


Figure 1: Circuit architecture of the VCH–T4/T5 microcircuit. Numbers represent total synapse counts from the 918 reciprocal T4/T5 population.

1. **T4/T5 is the dominant input to VCH** (47.2% of synapses), with balanced ON (T4a) and OFF (T5a) pathway convergence.
2. **VCH provides reciprocal feedback inhibition** to 89% of its T4/T5 inputs—a near-universal feedback loop characteristic of divisive normalization.
3. **The 918 reciprocal T4/T5 form a convergent-divergent hub**, broadcasting to 145,000 targets with LPi14 as the dominant synaptic sink.
4. **Network-level consequences:** T4/T5 motion signals are integrated by VCH, which provides proportional inhibitory feedback to the same T4/T5 population. Simultaneously, T4/T5 broadcasts motion information to HS output cells, lobula plate intrinsic interneurons (LPi), centrifugal feedback neurons (DCH, VCH), and modulatory central complex targets (Nod).
5. **Functional interpretation:** The VCH–T4/T5 loop implements gain control on the motion detection pathway. Local motion signals (T4/T5) drive VCH, which normalizes T4/T5 output, preventing saturation while preserving relative motion signals for downstream HS cells and central complex navigation circuits.

All raw data, including complete partner lists with synapse counts, annotations, and spatial coordinates, are available in the accompanying JSON files at `/tmp/vch_v2_complete.json`, `/tmp/vch_left_t4t5_in` and `/tmp/vch_t4t5_downstream_targets.json`.

References

- [1] S. Dorkenwald *et al.*, Neuronal wiring diagram of an adult brain. *Nature* **634**, 124–138 (2024).
- [2] P. Schlegel *et al.*, Whole-brain annotation and multi-connectome cell typing of *Drosophila*. *Nature* **634**, 139–152 (2024).
- [3] M. S. Maisak *et al.*, A directional tuning map of *Drosophila* elementary motion detectors. *Nature* **500**, 212–216 (2013).
- [4] FlyWire Consortium, *Codex: Connectome Data Explorer*. <https://codex.flywire.ai>

- [5] CAVEclient Python package, Seung Lab, Princeton University. <https://github.com/seung-lab/CAVEclient>